

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

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## Device Specifications

Device Type: Video Projector  
Manufacturer: Casio  
Firmware Version: N/A  
Model(s): XJ-L8300HN

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 3.01.0000-b010                         | 2.1.0                   |

## Version History

| Module Version | Date     | Notes                         |
|----------------|----------|-------------------------------|
| 1_1_0_0        | 1/2/2019 | Added PJLink Ethernet Control |
| 1_0_0_0        | 2/6/2018 | Initial Version               |

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## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required devicePassword must be set accordingly. devicePassword default value is 'admin'.  
Example: `InterfaceName.devicePassword = 'admin'`

## Supported Classes and Examples

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                                           |
| <code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='XJ-L8300HN')</code>               |
| <b>SerialOverEthernetClass</b>                                                                               |
| <code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='XJ-L8300HN')</code> |
| <b>EthernetClass</b>                                                                                         |
| <code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='XJ-L8300HN')</code>           |

## Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                   |                                          |                            |                             |
|-------------------------------------------------------------------|------------------------------------------|----------------------------|-----------------------------|
| Command<br><b>AspectRatio</b>                                     | Value<br>'4:3'<br>'Auto'                 | Value<br>'16:9'            | Value<br>'16:10'            |
| # AspectRatio example<br>InterfaceName.Set('AspectRatio', '4:3')  |                                          |                            |                             |
| Command<br><b>AudioMute</b>                                       | Value<br>'On'                            | Value<br>'Off'             |                             |
| # AudioMute example<br>InterfaceName.Set('AudioMute', 'On')       |                                          |                            |                             |
| Command<br><b>AutoImage</b>                                       | Value<br>None                            |                            |                             |
| # AutoImage example<br>InterfaceName.Set('AutoImage', None)       |                                          |                            |                             |
| Command<br><b>Input</b>                                           | Value<br>'Computer / YPbPr'<br>'HDBaseT' | Value<br>'HDMI 1'          | Value<br>'HDMI 2'           |
| # Input example<br>InterfaceName.Set('Input', 'Computer / YPbPr') |                                          |                            |                             |
| Command<br><b>KeyLock</b>                                         | Value<br>'On'                            | Value<br>'Off'             |                             |
| # KeyLock example<br>InterfaceName.Set('KeyLock', 'On')           |                                          |                            |                             |
| Command<br><b>LampMode</b>                                        | Value<br>'Normal'                        | Value<br>'Eco'             | Value<br>'360 and Portrait' |
| # LampMode example<br>InterfaceName.Set('LampMode', 'Normal')     |                                          |                            |                             |
| Command<br><b>MenuCall</b>                                        | Value<br>'On'                            | Value<br>'Off'             |                             |
| # MenuCall example<br>InterfaceName.Set('MenuCall', 'On')         |                                          |                            |                             |
| Command<br><b>MenuNavigation</b>                                  | Value<br>'Up'<br>'Right'                 | Value<br>'Down'<br>'Enter' | Value<br>'Left'<br>'Back'   |
| # MenuNavigation example                                          |                                          |                            |                             |

## Global Scripter Module Communication Sheet

|                                                             |               |                 |
|-------------------------------------------------------------|---------------|-----------------|
| InterfaceName.Set('MenuNavigation', 'Up')                   |               |                 |
| Command<br><b>Power</b>                                     | Value<br>'On' | Value<br>'Off'  |
| # Power example<br>InterfaceName.Set('Power', 'On')         |               |                 |
| Command<br><b>VideoMute</b>                                 | Value<br>'On' | Value<br>'Off'  |
| # VideoMute example<br>InterfaceName.Set('VideoMute', 'On') |               |                 |
| Command<br><b>Volume</b>                                    | Value<br>'Up' | Value<br>'Down' |
| # Volume example<br>InterfaceName.Set('Volume', 'Up')       |               |                 |

## Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                         |                             |                   |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------|-------------------|
| Command<br><b>AspectRatio</b>                                                                                                                                                           | Value<br>'4:3'              | Value<br>'16:9'   | Value<br>'16:10'  |
|                                                                                                                                                                                         | 'Auto'                      |                   |                   |
| # AspectRatio examples<br>InterfaceName.Update('AspectRatio')<br>Value = InterfaceName.ReadStatus('AspectRatio')<br>InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler) |                             |                   |                   |
| Command<br><b>AudioMute</b>                                                                                                                                                             | Value<br>'On'               | Value<br>'Off'    |                   |
| # AudioMute examples<br>InterfaceName.Update('AudioMute')<br>Value = InterfaceName.ReadStatus('AudioMute')<br>InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)         |                             |                   |                   |
| Command<br><b>Input</b>                                                                                                                                                                 | Value<br>'Computer / YPbPr' | Value<br>'HDMI 1' | Value<br>'HDMI 2' |
|                                                                                                                                                                                         | 'HDBaseT'                   |                   |                   |
| # Input examples<br>InterfaceName.Update('Input')<br>Value = InterfaceName.ReadStatus('Input')<br>InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)                         |                             |                   |                   |
| Command<br><b>KeyLock</b>                                                                                                                                                               | Value<br>'On'               | Value<br>'Off'    |                   |
| # KeyLock examples<br>InterfaceName.Update('KeyLock')<br>Value = InterfaceName.ReadStatus('KeyLock')<br>InterfaceName.SubscribeStatus('KeyLock', None, FeedbackHandler)                 |                             |                   |                   |
| Command                                                                                                                                                                                 | Value                       | Value             | Value             |

## Global Scripter Module Communication Sheet

Revision: 1/2/2019

|                                                                                                                                                                                   |                |                |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|--------------------|
| <b>LampMode</b>                                                                                                                                                                   | 'Normal'       | 'Eco'          | '360 and Portrait' |
| <pre># LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>     |                |                |                    |
| <b>Command</b><br><b>LampUsage</b>                                                                                                                                                | Value<br>Hours |                |                    |
| <pre># LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre> |                |                |                    |
| <b>Command</b><br><b>Power</b>                                                                                                                                                    | Value<br>'On'  | Value<br>'Off' |                    |
| <pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>                 |                |                |                    |
| <b>Command</b><br><b>VideoMute</b>                                                                                                                                                | Value<br>'On'  | Value<br>'Off' |                    |
| <pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre> |                |                |                    |

## Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                   |                                          |                               |
|-------------------------------------------------------------------|------------------------------------------|-------------------------------|
| Command<br><b>AudioMute</b>                                       | Value<br>'On'                            | Value<br>'Off'                |
| # AudioMute example<br>InterfaceName.Set('AudioMute', 'On')       |                                          |                               |
| Command<br><b>AVMute</b>                                          | Value<br>'On'                            | Value<br>'Off'                |
| # AVMute example<br>InterfaceName.Set('AVMute', 'On')             |                                          |                               |
| Command<br><b>Input</b>                                           | Value<br>'Computer / YPbPr'<br>'HDBaseT' | Value<br>'HDMI 1'<br>'HDMI 2' |
| # Input example<br>InterfaceName.Set('Input', 'Computer / YPbPr') |                                          |                               |
| Command<br><b>Power</b>                                           | Value<br>'On'                            | Value<br>'Off'                |
| # Power example<br>InterfaceName.Set('Power', 'On')               |                                          |                               |
| Command<br><b>VideoMute</b>                                       | Value<br>'On'                            | Value<br>'Off'                |
| # VideoMute example<br>InterfaceName.Set('VideoMute', 'On')       |                                          |                               |

## Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                             |                             |                              |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------|-----------------------|
| <b>Command</b><br><b>AudioMute</b>                                                                                                                                                          | Value<br>'On'               | Value<br>'Off'               |                       |
| # AudioMute examples<br>InterfaceName.Update('AudioMute')<br>Value = InterfaceName.ReadStatus('AudioMute')<br>InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)             |                             |                              |                       |
| <b>Command</b><br><b>AVMute</b>                                                                                                                                                             | Value<br>'On'               | Value<br>'Off'               |                       |
| # AVMute examples<br>InterfaceName.Update('AVMute')<br>Value = InterfaceName.ReadStatus('AVMute')<br>InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)                         |                             |                              |                       |
| <b>Command</b><br><b>DeviceStatus</b>                                                                                                                                                       | Value<br>'Normal'           | Value<br>'Fan Warning'       | Value<br>'Fan Error'  |
|                                                                                                                                                                                             | 'Lamp Warning'              | 'Lamp Error'                 | 'Temperature Warning' |
|                                                                                                                                                                                             | 'Temperature Error'         | 'Cover Warning'              | 'Cover Error'         |
|                                                                                                                                                                                             | 'Filter Warning'            | 'Filter Error'               | 'Other Warning'       |
|                                                                                                                                                                                             | 'Other Error'               | 'Multiple Warnings / Errors' |                       |
| # DeviceStatus examples<br>InterfaceName.Update('DeviceStatus')<br>Value = InterfaceName.ReadStatus('DeviceStatus')<br>InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler) |                             |                              |                       |
| <b>Command</b><br><b>Input</b>                                                                                                                                                              | Value<br>'Computer / YPbPr' | Value<br>'HDMI 1'            | Value<br>'HDMI 2'     |
|                                                                                                                                                                                             | 'HDBaseT'                   |                              |                       |

|                                                                                                                                                                                   |                |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| <pre># Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>                 |                |                |
| <b>Command</b><br><b>LampUsage</b>                                                                                                                                                | Value<br>Hours |                |
| <pre># LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre> |                |                |
| <b>Command</b><br><b>Power</b>                                                                                                                                                    | Value<br>'On'  | Value<br>'Off' |
| <pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>                 |                |                |
| <b>Command</b><br><b>VideoMute</b>                                                                                                                                                | Value<br>'On'  | Value<br>'Off' |
| <pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre> |                |                |

## Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

## Notes for the Device

## Serial communication

**Port Type:** RS-232

**Baud Rate:** 9600

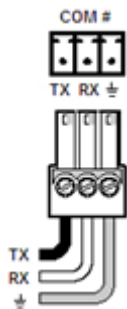
**Data Bits:** 8

**Parity:** None

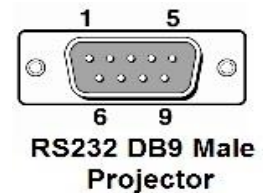
**Stop Bits:** One

**Flow Control:** None

## Pin Assignments Diagram



| Signal | Main Cable | Pin | Signal |
|--------|------------|-----|--------|
| TxD    | →          | 2   | RxD    |
| RxD    | ←          | 3   | TxD    |
| GND    | →          | 5   | GND    |



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## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                                     |          |
|-------------------------------------|----------|
| <b>Port Type:</b>                   | Ethernet |
| <b>Default Port:</b>                | 4352     |
| <b>Logon Credentials Supported:</b> | Yes      |
| <b>Default Password:</b>            | admin    |
| <b>Multi-Connection</b>             | Yes      |
| <b>Capabilities:</b>                |          |
| <b>Port Changeability:</b>          | No       |

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## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device

The PJLink Connection Authentication is always enabled with this device. There is no option to disable it from Menu.

**Appendix A. Set Commands (Serial)**

|                                   |                             |
|-----------------------------------|-----------------------------|
| <b>Aspect Ratio 16:10</b>         | \x0D*asp=16:10#\x0D         |
| <b>Aspect Ratio 16:9</b>          | \x0D*asp=16:9#\x0D          |
| <b>Aspect Ratio 4:3</b>           | \x0D*asp=4:3#\x0D           |
| <b>Aspect Ratio Auto</b>          | \x0D*asp=AUTO#\x0D          |
| <b>Audio Mute Off</b>             | \x0D*mute=off#\x0D          |
| <b>Audio Mute On</b>              | \x0D*mute=on#\x0D           |
| <b>Auto Image None</b>            | \x0D*auto#\x0D              |
| <b>Input Computer / YPbPr</b>     | \x0D*sour=RGB#\x0D          |
| <b>Input HDBaseT</b>              | \x0D*sour=hdbaset#\x0D      |
| <b>Input HDMI 1</b>               | \x0D*sour=hdmi#\x0D         |
| <b>Input HDMI 2</b>               | \x0D*sour=hdmi2#\x0D        |
| <b>Key Lock Off</b>               | \x0D*keylock=off#\x0D       |
| <b>Key Lock On</b>                | \x0D*keylock=on#\x0D        |
| <b>Lamp Mode 360 and Portrait</b> | \x0D*lampm=portrait360#\x0D |
| <b>Lamp Mode Eco</b>              | \x0D*lampm=eco#\x0D         |
| <b>Lamp Mode Normal</b>           | \x0D*lampm=lnor#\x0D        |
| <b>Menu Call Off</b>              | \x0D*menu=off#\x0D          |
| <b>Menu Call On</b>               | \x0D*menu=on#\x0D           |
| <b>Menu Navigation Back</b>       | \x0D*back#\x0D              |
| <b>Menu Navigation Down</b>       | \x0D*down#\x0D              |
| <b>Menu Navigation Enter</b>      | \x0D*enter#\x0D             |
| <b>Menu Navigation Left</b>       | \x0D*left#\x0D              |
| <b>Menu Navigation Right</b>      | \x0D*right#\x0D             |
| <b>Menu Navigation Up</b>         | \x0D*up#\x0D                |
| <b>Power Off</b>                  | \x0D*pow=off#\x0D           |
| <b>Power On</b>                   | \x0D*pow=on#\x0D            |
| <b>Video Mute Off</b>             | \x0D*blank=off#\x0D         |
| <b>Video Mute On</b>              | \x0D*blank=on#\x0D          |
| <b>Volume Down</b>                | \x0D*vol=-#\x0D             |
| <b>Volume Up</b>                  | \x0D*vol=+#\x0D             |

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## Appendix B. Update Commands (Serial)

|                     |                     |
|---------------------|---------------------|
| <b>Aspect Ratio</b> | \x0D*asp=?#\x0D     |
| <b>Audio Mute</b>   | \x0D*mute=?#\x0D    |
| <b>Input</b>        | \x0D*sour=?#\x0D    |
| <b>Key Lock</b>     | \x0D*keylock=?#\x0D |
| <b>Lamp Mode</b>    | \x0D*lampm=?#\x0D   |
| <b>Lamp Usage</b>   | \x0D*ltim=?#\x0D    |
| <b>Power</b>        | \x0D*pow=?#\x0D     |
| <b>Video Mute</b>   | \x0D*blank=?#\x0D   |

## Appendix C. Set Commands (Ethernet)

|                               |               |
|-------------------------------|---------------|
| <b>Audio Mute Off</b>         | %1AVMT 20\x0D |
| <b>Audio Mute On</b>          | %1AVMT 21\x0D |
| <b>AV Mute Off</b>            | %1AVMT 30\x0D |
| <b>AV Mute On</b>             | %1AVMT 31\x0D |
| <b>Input Computer / YPbPr</b> | %1INPT 11\x0D |
| <b>Input HDBaseT</b>          | %1INPT 51\x0D |
| <b>Input HDMI 1</b>           | %1INPT 31\x0D |
| <b>Input HDMI 2</b>           | %1INPT 32\x0D |
| <b>Power Off</b>              | %1POWR 0\x0D  |
| <b>Power On</b>               | %1POWR 1\x0D  |
| <b>Video Mute Off</b>         | %1AVMT 10\x0D |
| <b>Video Mute On</b>          | %1AVMT 11\x0D |

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## Appendix D. Update Commands (Ethernet)

|                      |              |
|----------------------|--------------|
| <b>AV Mute</b>       | %1AVMT ?\x0D |
| <b>Device Status</b> | %1ERST ?\x0D |
| <b>Input</b>         | %1INPT ?\x0D |
| <b>Lamp Usage</b>    | %1LAMP ?\x0D |
| <b>Power</b>         | %1POWR ?\x0D |